

KATHMANDU UNIVERSITY

Course of Study

Course Code: DSMA 226

Credit Hours: 3

Course Title: Multivariate Analysis

Teaching Hours: 45

Group: Bachelor of Data Science (II Year - II Semester)

Course Description

This course provides an introduction to the fundamental concepts and practical techniques of multivariate data analysis within the context of Data Science. It focuses on analyzing datasets containing multiple variables simultaneously, emphasizing data visualization, dimensionality reduction, clustering, classification, feature selection, and multivariate hypothesis testing. Students learn to explore and interpret high-dimensional data, reduce complexity using methods such as Principal Component Analysis (PCA), identify patterns and group structures, and apply basic supervised and unsupervised learning approaches. Practical sessions using statistical software such as R or Python are integrated throughout the course to develop hands-on skills in implementation, interpretation, and reporting, with greater emphasis on applied analysis and real-world data problems than on mathematical derivations.

Course Objectives

The main objective of this course is to provide students with foundational knowledge and practical skills in multivariate data analysis, enabling them to explore, analyze, and interpret high-dimensional datasets using appropriate statistical techniques and computational tools. More specifically, this course aims to:

- (a)** Familiarize students with the structure and characteristics of multivariate data, including covariance, correlation, and data visualization techniques.
- (b)** Develop students' ability to apply dimensionality reduction methods such as Principal Component Analysis (PCA) for feature extraction and data simplification.
- (c)** Equip learners to perform clustering techniques for identifying patterns and group structures in data.
- (d)** Enable students to implement basic supervised classification methods and feature selection techniques for predictive modeling.
- (e)** Develop understanding and interpretation of factor analysis for identifying latent structures in multivariate datasets.
- (f)** Enable students to apply introductory multivariate testing procedures, including MANOVA and Hotelling's T^2 , and interpret their results using statistical software.

Course Content & Syllabus

UNIT 1: Introduction to Multivariate Analysis (6 LH)

- Meaning and types of multivariate data, Big data, Multivariate analysis and its implications
- Measurement scales and data processing, measurement errors
- Basic matrix algebra: vectors and matrices, transpose and inverse of a matrix, eigenvalues and eigenvectors
- Multivariate marginal and conditional distributions; means, variances, covariance and correlation matrices, linear combination of variables
- Visualization Techniques: scatterplots, pairwise plots, correlation heatmaps, biplots
- Bivariate and multivariate normal distribution: concepts, properties, and QQ-plots
- Computer Application: Computing mean vector, covariance and correlation matrix and scatter plot matrix

UNIT 2: Principal Component Analysis (PCA) (8 LH)

- Curse of dimensionality and dimension reduction, Standardization and scaling, Idea of principal components, Variance explained and scree plot, Interpretation of components loadings
- Principal components using covariance / correlation matrix, PCA for visualization and feature reduction, Dimensionality reduction beyond PCA
- Applications in data compression and pattern detection: Performing PCA in R/Python, Generating Scree plot, interpretation of output, Reducing features using PCA

UNIT 3: Cluster Analysis (6 Hours)

- Concept of clustering, Distance measures (Euclidean distance – concept only)
- K-means clustering, Hierarchical clustering
- Dendrogram interpretation and choosing the optimal number of clusters
- Applications in segmentation and profiling

UNIT 4: Factor Analysis (6 LH)

- Concept of latent variables, Introduction to Factor Analysis
- Difference between PCA and Factor Analysis
- Basic Factor models (conceptual form only), Factor loadings and communalities (interpretation only)
- Factor Rotation (Varimax – concept only)
- Applications in survey analysis

UNIT 5: Classification Methods and Feature Selection Techniques (7 LH)

- Supervised vs unsupervised learning
- Correlation-based selection

- Variance threshold
- Multivariate Regression and Review of logistic regression (brief)
- Linear Discriminant Analysis (conceptual approach)
- Confusion matrix and performance matrix

UNIT 6: Introduction to Multivariate Testing (9 Hours)

- Review of ANOVA, Need for MANOVA, One way MANOVA
- Hotelling's T^2 (concept only)
- Interpretation of Wilks' Lambda
- Assumptions (multivariate normality, homogeneity) and diagnostics
- Real data example

References

- Everitt, B. S., & Hothorn, T. (2011). An Introduction to Applied Multivariate Analysis with R. Springer.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning: With Applications in R (2nd ed.). Springer.
- Johnson, R. A., & Wichern, D. W. (2018). Applied Multivariate Statistical Analysis (6th ed.). Pearson.
- Rencher, A. C., & Christensen, W. F. (2012). Methods of Multivariate Analysis (3rd ed.). Wiley.
- Scott, D. W. (2015). Multivariate Density Estimation: Theory, Practice, and Visualization (2nd ed.). Wiley.

Evaluation Modality

1. In-Semester Evaluation: 40 marks

2. End Semester Evaluation: 60 marks

Category	Weightage	Format & Breakdown
Objective Questions	20 marks	10 Q. $\times$ 1 = 10 marks for Fill in the Blank questions 10 Q. $\times$ 1 = 10 marks for Multiple Choice questions
Subjective Questions	40 marks	2 Q. $\times$ 8 = 16 marks for Long Answer questions 6 Q. $\times$ 4 = 24 marks for Short Answer questions